

INDEPENDENT PEER-REVIEW PACKET

EC-011

Adjoint-Orbit Conservation and Section Normalization

Cold Review Packet 6 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	6 of 20
Object ID	EC-011
Object class	Exact differential identity
Exact statement	For $\gamma_{\dot{}} = \text{grad } f$ and $w_{\dot{}} = -H_f w$ with symmetric H_f , $d/dt[w \cdot \text{grad } f(\gamma)] = 0$. The value is identically zero only when the declared terminal transverse-section normalization sets it to zero.
Internal terminal label	CLOSED - EXACT ADJOINT-ORBIT CONSERVATION / SECTION-NORMALIZATION IDENTITY
Terminal date	2026-07-21
Recommended expertise	Dynamical systems / differential geometry / variational equations

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Direct chain-rule cancellation with a separate normalization consequence.
Prior review record	Independent derivation verified conservation and fenced it from the unresolved global transversality theorem.
Binding exclusions	No global Morse-Smale transversality, charting, or probability conclusion.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.
- Object-specific exclusions: No global Morse-Smale transversality, charting, or probability conclusion.

4. Object-specific hostile checks

- Differentiate the pairing using the two ODEs.
- Identify exactly where Hessian symmetry is used.
- Construct a nonzero initial pairing to show conservation does not imply zero without normalization.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

GP-CLS-PROP-014-v1.0 — PROPOSED TERMINAL CLOSURE OF THE ADJOINT-ORBIT CONSERVATION AND SECTION-NORMALIZATION IDENTITY

Artifact ID: GP-CLS-PROP-014-v1.0

Author: OpenAI GPT-5.6 Thinking

Created: 2026-07-21

Class: CLS-PROP — additive terminal-closure proposal

Status: READY FOR HUMAN APPROVAL

Authority: none until Dylan M. Roy approves or modifies the terminal statement

Canonical impact: NONE before approval

1. Exact object card

OBJECT CLASS: DETERMINISTIC DIFFERENTIAL IDENTITY ALONG A GRADIENT ORBIT

FIELD: a C^2 scalar field f on a Riemannian two-dimensional chart, expressed in Euclidean local coordinates for the displayed calculation

ORBIT: a C^1 curve γ satisfying $\dot{\gamma}(t) = \nabla f(\gamma(t))$

ADJOINT: a C^1 vector/covector representative w satisfying $\dot{w}(t) = -H_f(\gamma(t))w(t)$

SYMMETRY: H_f is symmetric

TERMINAL NORMALIZATION: $w(t_+) \cdot \dot{\gamma}(t_+) = 0$ at the chosen transverse section

EVIDENCE CLASS: exact differentiation plus section-normalization implication

PROPOSED TERMINAL LABEL: CLOSED — EXACT ADJOINT-ORBIT CONSERVATION / SECTION-NORMALIZATION IDENTITY

2. Exact terminal statements

Conservation statement:

For every t on the regular orbit segment,

$$d/dt [w(t) \cdot \nabla f(\gamma(t))] = 0.$$

Therefore $w \cdot \nabla f$ is constant along the orbit.

Normalization consequence:

If the terminal section convention imposes

$$w(t_+) \cdot \dot{\gamma}(t_+) = 0,$$

then, because $\dot{\gamma} = \nabla f$,

$$w(t) \cdot \dot{\gamma}(t) = w(t) \cdot \nabla f(\gamma(t)) = 0$$

for every t on the same orbit segment.

Thus the adjoint direction is everywhere normal to the gradient orbit under the declared normalization.

3. Fresh second-line derivation

Using the product and chain rules,

$$\begin{aligned} \frac{d}{dt} [w \cdot \nabla f(\gamma)] \\ = \dot{w} \cdot \nabla f(\gamma) + w \cdot [H_f(\gamma) \dot{\gamma}]. \end{aligned}$$

Substitute the declared equations $\dot{w} = -H_f w$ and $\dot{\gamma} = \nabla f$:

$$= (-H_f w) \cdot \nabla f + w \cdot (H_f \nabla f).$$

Since H_f is symmetric,

$$(-H_f w) \cdot \nabla f = -w \cdot H_f \nabla f.$$

Hence the two terms cancel exactly and the derivative is zero.

This proves constancy. It does not prove that the constant is zero without a normalization condition. Under the declared transverse-section normalization,

$$w(t_+) \cdot \dot{\gamma}(t_+) = 0,$$

so the constant is zero at t_+ and therefore zero throughout the orbit segment.

4. Meaning of the section-normalization step

At the terminal crossing, displacement tangent to the flow changes the crossing time rather than the location of the crossing on the fixed transverse section. The mismatch derivative is represented in the section-normal direction. The adjoint terminal condition therefore annihilates the tangent vector $\dot{\gamma}(t_+)$.

This proposal closes only the exact implication once that section convention is adopted. It does not independently construct the chart, prove that the crossing is regular, or prove the global measurability and disintegration arguments.

5. Evidence and review classification

Primary source: AI edit 0004 — 02_GAUSSIAN_TRANSVERSALITY.md, Review Note 2, Drive ID 1znQn1YWxcR3yPw-DtJSyWLThAGU_0JOc.

Closure census: AO48-REG-018-v1.0 identified the section-normalization check as the sole remaining second-line step for S5.

Second-line route completed here: OpenAI reconstructed the differentiation from the primitive gradient-flow and adjoint equations and separated the unconditional conservation statement from the normalization-dependent zero statement.

Independence classification: strong dependent verification. The source identity was visible, but the calculation was independently reconstructed line by line. This is sufficient for narrow deterministic closure, not external-referee independence for the surrounding theorem.

6. Scope boundary and forbidden extrapolations

This proposed closure establishes only:

- constancy of $w \cdot \nabla f$ along the declared gradient/adjoint orbit;
- zero constancy under the declared terminal section normalization;
- the resulting normality of w to the orbit under that normalization.

It does not by itself establish:

- existence or smooth tracking of the saddle pair or orbit chart;
- correctness of all endpoint atom formulas;
- nonvanishing of every scalar adjoint weight used later;
- a nonzero curve contribution without the normal-bump and weight assumptions;
- countable chart coverage;
- uniform lower bounds on eigenvalue ratios;
- measurable selection or one-dimensional disintegration;
- exclusion of exceptional shift parameters;
- the almost-sure Morse–Smale/SARD-G theorem;
- any rate theorem, P0.2, Theorem B, or machine-root transition.

The surrounding Gaussian transversality theorem remains uncertain and must not be described as closed by this identity.

7. Dependencies affected

If approved, this closure:

- repairs the specific local objection that continuity and nonvanishing alone did not exclude a tangential adjoint contribution;
- permits exact citation that the normalized adjoint is normal to the orbit;
- supports the later normal-coordinate bump construction subject to its remaining hypotheses;
- retires repeated differentiation and normalization checks for this exact object.

8. Dependencies explicitly unaffected

Unaffected:

- all global charting, covering, slicing, exceptional-parameter, uniformity, and disintegration obligations;
- the status of the Morse–Smale/R0 theorem;
- the conditional status of any theorem assuming Morse–Smale;
- P0.2, Theorem B, canonical theorem status, and active machine state.

9. Coupled-advancement architecture response

Mathematical status change proposed: one deterministic conservation identity and one explicitly normalized consequence move from a self-flagged review note to second-line-verified narrow closure.

Architectural response required: retain separate fields for “conservation” and “zero under normalization.” This prevents the zero conclusion from being cited without the terminal condition and prevents a local identity from laundering the global theorem.

No broader architecture change is required because the existing object-card, dependency, scope, and terminal-closure controls already represent this transition.

Failure that would reveal this judgment was insufficient:

- a later source drops the normalization but still asserts $w \cdot \nabla f = 0$;
- the adjoint sign convention or flow direction is changed without rechecking;

- the closed local identity is cited as closure of Morse–Smale/R0.

10. Human approval requirement

Human approval is REQUIRED because terminal mathematical closure remains reserved to Dylan M. Roy.

Suggested approval statement:

“I approve GP-CLS-PROP-014-v1.0 exactly as a closure of the declared adjoint-orbit conservation identity and its zero consequence under the stated terminal section normalization. Record the terminal label CLOSED — EXACT ADJOINT-ORBIT CONSERVATION / SECTION-NORMALIZATION IDENTITY, preserve all provenance, and do not infer closure of Proposition 6.1 in full, Morse–Smale/R0, any rate theorem, P0.2, Theorem B, or machine state.”

11. Additive correction and reopening path

Reopening requires new contradictory evidence identifying a sign-convention error, a failure of Hessian symmetry in the declared setting, a mismatch in the terminal section normalization, or a misuse of the identity beyond its scope. Corrections must preserve this proposal and primary sources, map affected and unaffected dependencies, and reopen only this narrow object unless broader contamination is demonstrated.

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT ADJOINT-ORBIT CONSERVATION / SECTION-NORMALIZATION IDENTITY
Closure date	2026-07-21
Evidence basis	Direct chain-rule cancellation with a separate normalization consequence.
Independent review already recorded	Independent derivation verified conservation and fenced it from the unresolved global transversality theorem.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- No global Morse-Smale transversality, charting, or probability conclusion.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____
Reproduction artifacts	Code/log/hash/location: _____

Signature and date _____

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: https://docs.google.com/document/d/1HHQAwZCcxm_wDcMBFpzP-NLJD6xdGLxJ0ycrrck66yY/edit

Closure / terminal record: https://docs.google.com/document/d/1Zm65j9BZe_z3P_8qP7rfZukRU-IWUE8h8bDqXdDsIQ/edit

Local source snapshot used to construct this packet: GP-CLS-PROP-014-v1.0 — Proposed terminal closure of the adjoint-orbit conservation and section-normalization identity

Local source snapshot SHA-256 (DOCX export, transport checksum only):
ed31f957de38a8b86ccb775a48e7ac6edea86076507c20e53acd4e64009a3a35

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.